Lucas w. Davis
Paul Gertler
Ernesto Rodriguez

Wine Experts Defied

A New Wine Cellar

It is mid March, 1989. You are Manita DuBois, the new procurement officer at London-based Fûts de Chêne Rouge (FCR), a premium wine shop chain. The company is scheduled to open an exclusive wine cellar in Tokyo in 1991, exactly like the ones being built in Manhattan and Paris this year. Your first big project at FCR is to select the best wines available to stock this cellar, especially those coming from the world's most renowned wine countries. You are thinking of sourcing 50% of wines from the Bordeaux region in France.

In your search for industry sources, your team comes across traditional publications such as *The Wine Spectator*, *The Wine Advocate* and *International Wine Cellar*. While you do not know much about wine, you think you and your team can obtain significant insights from these publications, as well as data available from wine auctions. Your goal is to select wines representing a mix of premium and super-premium wines, (wholesale prices per bottle range from $30 to $200), which will be sold at high margins. It is a common practice in the industry to buy young wines and store them for 10 years–until they reach the maturity peak–before releasing them to the market.

During your search, you also find a publication from an unusual source: Lucas Mann, a Chicago-based commodity researcher, who periodically writes about wine prices in his *Red and White Assets* magazine. He is not exactly a wine expert, but has attracted a small group of followers that pay attention to his statistics-based predictions on Bordeaux wine prices.

A Long Winemaking Tradition

Bordeaux is the foremost name in the world for red wines. Though most wines are bottled soon after they're made and are ready to drink shortly thereafter, Bordeaux wines spend 18 to 24 months in oak casks and are then set aside for aging in bottles. As the wine matures, it loses its initial astringency while flavors and aromas develop completely. Tastes from the barrel heavily influence the prices of these initial sales – rarely less than $30 a bottle – at which point the wines are shipped around the world. Experts say only years later can vintages be ranked by taste with any precision. And as most of them acknowledge, there is many a slip between barrel, bottle and lip.

Bordeaux is located in Western France crossed by the Gironde estuary, giving an oceanic climate to the whole region (see Exhibit A). The earliest evidence of winemaking in the region goes back to the Roman era (70 AD). Although it was popular locally, the Bordeaux wine was not exported until the 12th century, when the region was annexed by the English crown. Henry II of England abolished import taxes on Bordeaux wines, thus causing an expansion of wine production in this region. During the 13th and 14th centuries the wines from this region were given a distinct trade advantage over other regions, contributing to the expansion of wine production.

In 1725, the spread of vineyards throughout Bordeaux was so vast that it was divided into specific areas so that the consumer could tell exactly where each wine was from. The collection of districts was known as the Vignoble de Bordeaux, and bottles were labeled with both the region and the area from which they originated. To prevent other winemakers from labeling their wines with the Bordeaux brand, the French government passed a law in 1936 stating that all regions in France had to name their wines by the place in which they had been produced.

An Unusual Voice in a Glamorous Industry

Wine experts' opinions on the quality of different wines greatly affect the industry. Yet in spite of their power to influence market demand and prices, they are known to make an average living. Most experts say they cannot afford the premium wines they taste but they really love what they do – including having access to the most exclusive wines, getting invitations from famous winemakers, going on wine tasting trips and keeping samples for themselves. Consequently, wine experts see themselves as an exclusive club of gifted people.

When *Red and White Assets* first came to the eyes of wine experts they were infuriated. The magazine released a series of predictions on Bordeaux wine prices based on the weather conditions in the year the grapes were grown and harvested (see Exhibits B and C for the descriptive statistics and regression analysis from the published study). To experts, wine tasting is one of the most delicate skills human beings can develop; using numbers to identify quality wines kills the romance of tasting and turns wine

lovers who believe in those predictions into troglodytes. Moreover, wine experts were insulted that Lucas dared to write about the quality of wines that, by his own admission, he had never tasted.

Antono Ego, the most renowned French wine critic, was particularly infuriated. Antono has over 40 years in the winemaking industry and is related to the family that runs one of the most prestigious wineries in the Bordeaux region. His opinion on quality wines is regarded as the most accurate in the industry. Though Antono knew that weather affects the quality of wines (even for the same vineyard the quality of grapes is hard to predict), he claimed that no model could actually predict quality differences in Bordeaux wines. Antono also mentioned several flaws in Mann's approach:

> *"This is absurd; you cannot force wine quality to follow a formula. First of all, the human element in wine tasting cannot be replaced by any formula or algorithm. Secondly, wines are complex, very much like people: you have to know wines before making yourself an opinion of them. Mann's cold-calculations approach is insulting to the winemakers, critics and wine enthusiasts."*

Buy Low, Sell High

One of Manita's main concerns is to choose wisely among vintages from 1981 to 1988. Buying the wrong bottle can be maddening for consumers and catastrophic for FCR. Manita's team is focused on finding the finest wines on the market and avoiding mediocre vintages.

As a recent Haas MBA, Manita is more inclined to trust hard data than the sometimes contradictory opinions of experts. In addition, she thinks there is enough statistical power in Mann's regression to consider the study relevant. However, she is concerned that the magazine has less than 3% of the readership of the average established wine publication.

Experts have argued that the 1980s include some of the century's best Bordeaux vintages due to exceptionally warm weather and low precipitations during the growing season (see Exhibit D for Bordeaux weather data). Manita must focus on a few vintages that represent the highest quality wine. Should she select them based on the predictions of *Red and White Assets*? Or should she put her faith in the experts who've highlighted the 1982, 1985 and 1986 vintages?

Discussion Questions:

1. Who is Manita DuBois? What business problem is she facing?
2. How do wine experts assess quality?
3. What is Lucas Mann's approach?
4. Which vintages should Manita purchase? Why?

Exhibit A: Bordeaux Region in France

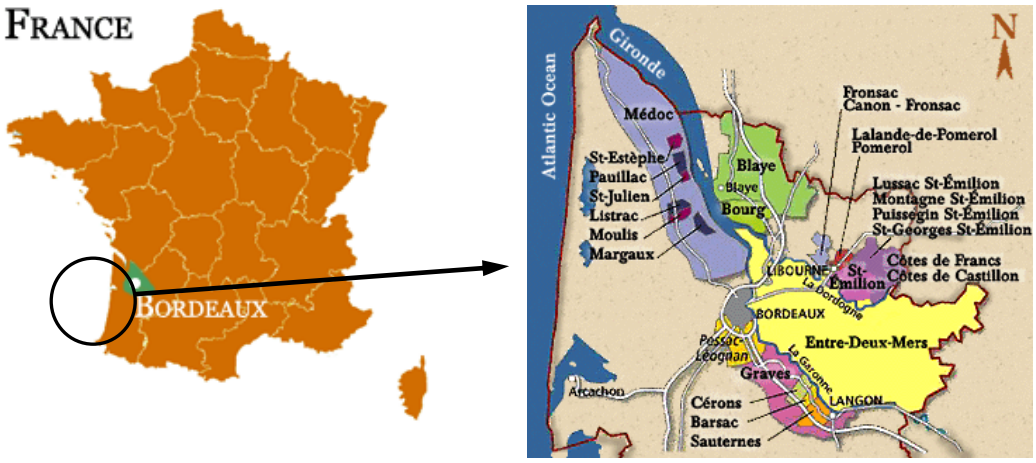


Exhibit B: Descriptive Statistics, Bordeaux Wines and Weather Variables

Variable	Mean	Standard Deviation	Minimum	Maximum
Price Per Bottle (in U.S. 1991 dollars)	$28.8	$21.0	$10.1	$100.0
Natural Log of Price	3.15	0.63	2.32	4.61
Age of Vintage (as of 1991)	24.2	8.2	11	39
Average Temperature over Growing Season from April to September (in degrees Celsius)	16.5	0.7	15.0	17.7
Rain During Harvest Season from August to September (in millimeters)	145	73	38	292
Rain in the Months Preceding the Growing Season from October to March (in millimeters)	608	129	376	830

Note: This table describes the vintages from 1952 to 1980, excluding the 1954 and 1956 vintages, which are now rarely sold. There are 27 total observations.

Exhibit C: Regression of the Price (in logs) on Weather Variables

Independent Variables	Estimate	Standard Error	*t*-statistic	p-value
Age of Vintage (as of 1991)	0.024	0.007	3.33	.003
Average Temperature over Growing Season from April to September (in degrees Celsius)	0.616	0.095	6.48	.000
Rain During Harvest Season from August to September (in millimeters)	-0.0039	0.0008	-4.78	.000
Rain in the Months Preceding the Growing Season from October to March (in millimeters)	0.0012	0.0005	2.42	.024
Intercept	-7.7	1.68	-4.61	.000

Note: All regressions use as data the vintages from 1952 to 1980, excluding the 1954 and 1956 vintages, which are now rarely sold. There are 27 total observations. The R^2 from the regression is 0.83 and the standard deviation of the residuals (s_e) is 0.29. The F-statistic for the model is 26.39 with a *p*-value of .0000.

Exhibit D: Bordeaux, France Weather Data for Years 1981 to 1988

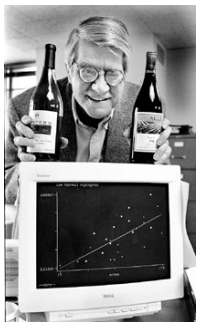
	Age (as of 1991)	Avg. Temperature over Growing Season (Apr-Sep)	Rain During Harvest Season (Aug-Sep)	Rain Preceding Season (Oct-March)
1981	10	17.0	111	535
1982	9	17.4	162	712
1983	8	17.4	119	830
1984	7	16.5	119	591
1985	6	16.8	38	744
1986	5	16.3	171	563
1987	4	17.0	115	452
1988	3	17.1	59	808

Note: All temperatures are in Celsius. Precipitation is measured in millimeters.

Exhibit E: Bordeaux Predictions, 1981-1988

	Prediction $\hat{y}$ (in logs)	95% Prediction Interval (in logs)	Prediction $\hat{y}$ (in dollars)	95% Prediction Interval	Picked By Experts
1981	3.22	2.64, 3.80	$25	$14, $45	No
1982	3.46	2.88, 4.04	$32	$18, $57	Yes
1983	3.76	3.18, 4.34	$43	$24, $77	No
1984	2.88	2.30, 3.46	$18	$10, $32	No
1985	3.54	2.96, 4.12	$34	$19, $62	Yes
1986	2.47	1.89, 3.05	$12	$7, $21	Yes
1987	2.96	2.38, 3.54	$19	$11, $35	No
1988	3.65	3.07, 4.23	$38	$21, $68	No

The prediction intervals were calculated using $\hat{y} \pm 2s_e$. See Stine and Foster, 2nd Edition, p. 553 for details.



Based On:

Ashenfelter, Orley, "Predicting the Quality and Prices of Bordeaux Wine" *Economic Journal*, 2008.

Ashenfelter, Orley, David Ashmore and Robert LaLonde, "Bordeaux Wine Vintage Quality and the Weather" *Chance*, 1995.

Passell, Peter, "Wine Equation Puts Some Noses Out of Joint" *New York Times*, March 4, 1990.